

Muskan Gupta

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EDUCATION

Vellore Institute of Technology, Chennai	2025 – 2027
Master of Computer Applications (MCA)	1st Semester: 86.2%
Khalsa College for Women, Punjab	July 2021 – June 2024
Bachelor of Computer Applications (BCA)	79.4%
Sawan Public Senior Secondary School, Punjab	2021
Senior Secondary (+2), PSEB	90.8%

SKILLS

Programming	: Python, Java, C
Data Analytics & Visualization	: Power BI, Tableau, Matplotlib, Excel
AI & ML	: TensorFlow, Keras, CNN, Deep Learning
Databases	: SQL, MongoDB
Web Development	: HTML, CSS
Tools	: NumPy, Pandas, Git

PROJECTS

- AI Mock Interview System** | *GenAI*
- Fine-tuned **TinyLlama** using **QLoRA** to build a scalable **interview question generator**.
 - Developed an **automated evaluation system** to **score responses** and provide **feedback**, enhancing mock interview experience.
 - Implemented pipeline using **HuggingFace ecosystem (Transformers, PEFT)**.
- CattleAI** | *AI-Powered Cattle Breed Recognition*
- Developed a CNN-based image classifier using **MobileNetV2** transfer learning (TensorFlow/Keras), achieving **85%+ accuracy** across multiple cattle breed categories.
 - Preprocessed and managed **large-scale image datasets** using NumPy/Pandas; reduced data loading time by **~40%** through pipeline optimizations and stored outputs in **MongoDB**.
- Crop Prediction** | *Machine Learning*
- Built a Scikit-learn ML model trained on **4 soil and climate features** to recommend optimal crops, achieving **~92% prediction accuracy**.
 - Performed feature engineering and hyperparameter tuning across **3+ ML algorithms**, improving model accuracy by **15%** over the baseline.
- Customer Behaviour & Shopping Trend Analysis** | *Python, SQL, Power BI*
- Conducted end-to-end **EDA** on **3,900+ records** using Python and **15+ SQL queries** to uncover **5+ purchasing patterns** and **3 high-value product categories** driving **60%+ of revenue**.
 - Built **3 interactive Power BI dashboards**, reducing manual reporting effort by **~60%** and delivering actionable business insights.
 - Managed multi-category datasets on **AWS S3** for scalable, reproducible analysis pipelines.

CERTIFICATIONS

- Neural Networks and Deep Learning** | *DeepLearning.AI* Mar 2026
- Completed **4-week** course; built neural networks from scratch and implemented backpropagation across **3+ deep learning architectures**.
- Deep Learning Specialization: CNNs** | *DeepLearning.AI* 2024 – 2025
- Implemented **3 CNN architectures** (LeNet, VGG, ResNet) for image classification; applied to real-world datasets with **85%+ accuracy**.
- Algorithms & Data Structures in Java** | *Udemy* Jan – Feb 2026
- Mastered **10+ data structures** (arrays, linked lists, stacks, queues, trees) and optimized algorithms, cutting average time complexity by **up to 50%** on practice problems.
- Deloitte Australia — Data Analytics Job Simulation** | *Forage* 2024 – 2025
- Cleaned and analyzed **real-world business datasets** across **2 simulation tasks**; generated insights that informed **3+ business recommendations** for a Fortune 500 client scenario.